# Veriscan-Cortex: Privacy-First Multi-Agent Fraud Detection & Intelligence Platform

## End-to-End Agentic AI System with Local LLM Inference, Behavioral Biometrics & Multimodal RAG


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## Abstract

- We present Veriscan-Cortex, an end-to-end privacy-first fraud detection and financial intelligence platform that combines hybrid ML modeling, multi-agent LLM orchestration, and multimodal retrieval-augmented generation (RAG) — all running entirely on local hardware with zero external API calls.
- The system processes 90,000+ transactions across 1,000 user archetypes through a 5-stage intelligence pipeline: Data Ingestion → Feature Engineering (19 signals) → Hybrid Fraud Scoring → Agentic AI Analysis → Identity Verification.
- Our multi-agent architecture achieves 90% tool-routing accuracy and 0.78 RAG Precision@3 while maintaining complete data sovereignty on Apple Silicon hardware via MLX-accelerated inference.


### Problem Statement:

Americans lose an estimated **$12.5 billion annually** to online fraud (Consumer Federation of America, 2024). Current fraud detection systems face three critical gaps:

**Privacy Risk:** Cloud-based AI systems (GPT-4o, Claude) require transmitting sensitive financial data to external servers, violating Tier-1 data privacy standards.

**Single-Model Limitation:** Traditional ML models (RF, XGBoost) detect fraud patterns but cannot provide contextual reasoning, financial advisory, or process multimodal evidence.

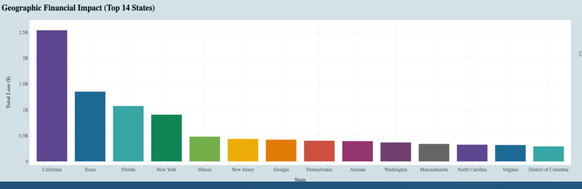
**Static Authentication:** Existing systems rely on one-time login credentials. They lack continuous behavioral identity verification that adapts to user spending patterns.

## Introduction

The global cost of cybercrime is projected to reach **$10.5 trillion by 2025** (Cybersecurity Ventures). Financial fraud, particularly in credit card and online shopping categories, accounts for the largest share of consumer losses. Investment scams and BEC (Business Email Compromise) lead global losses at **$6.57B** and **$2.7B** respectively (IC3 2024 Report).

Veriscan-Cortex addresses these challenges by introducing a **privacy-first, multi-agent AI architecture** that runs entirely on local Apple Silicon hardware. The system uniquely combines:

- **Hybrid ML + LLM fraud detection** with 19 engineered features
- **4 specialized AI agents** with tool-calling capabilities
- **8-axis behavioral biometric fingerprinting** (Spending DNA)
- **Privacy-preserving multimodal RAG** for document/image analysis
- **Zero-trust session isolation** across all intelligence layers

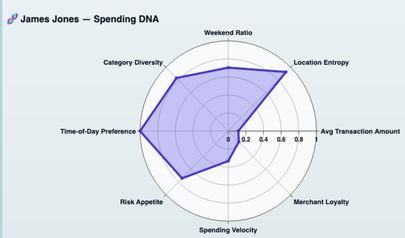


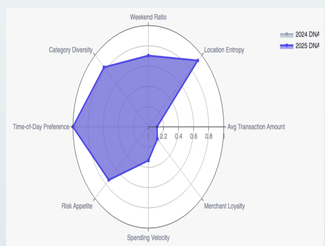

## Methodology

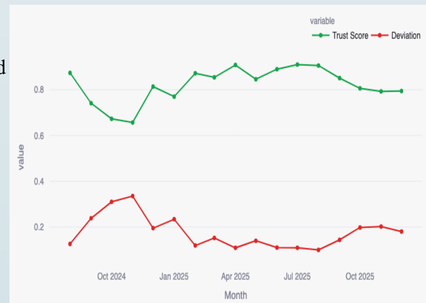
**Veriscan-Cortex follows a 5-stage intelligence pipeline:**

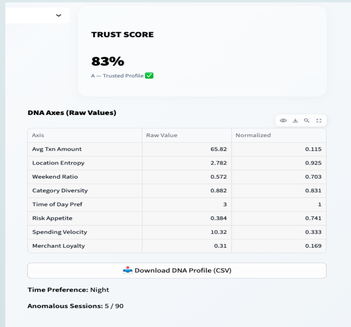
- **Data Ingestion & Synthesis:** 90,000 transactions generated across 1,000 user archetypes with 78% online fraud skew (matching real-world distribution). Data sources include the Kaggle fraud-detection dataset, IC3 2024 Report, and CFPB complaint database.
- **Feature Engineering (19 Signals):** Computed features including Amount Z-Score, transaction velocity (1h/24h/7d windows), location entropy, category risk weighting, weekend ratio, and temporal patterns.
- **Hybrid Fraud Modeling:** Random Forest classifier + heuristic risk scoring combined into COMBINED_RISK_SCORE with deterministic reproducibility (random_state=42) and 5× fraud oversampling.
- **Multi-Agent Agentic AI:** Four specialized agents — Security Analyst, Financial Advisor (with 3 sub-agents), Multimodal Intelligence, and Spending DNA — orchestrated via tool-calling routing.
- **Behavioral Identity (Spending DNA):** 8-axis behavioral fingerprinting computes trust scores by comparing baseline vs. session spending patterns for continuous authentication.

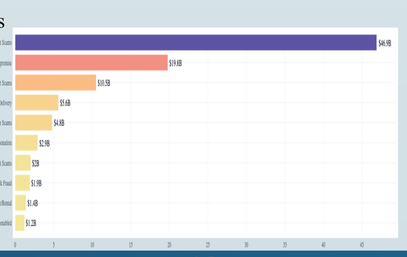
## Spending DNA

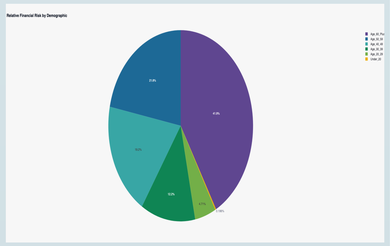


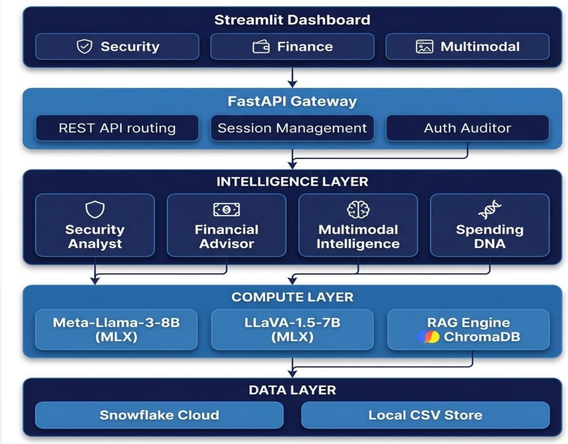



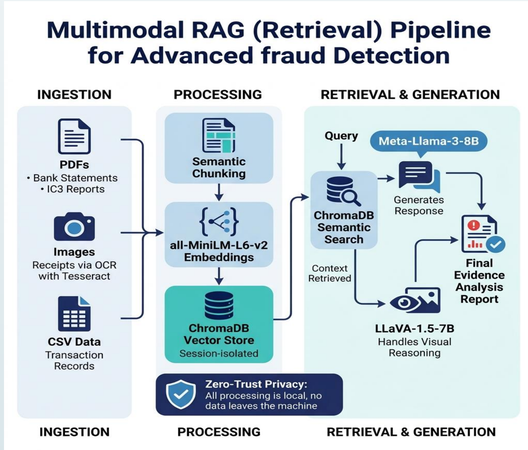



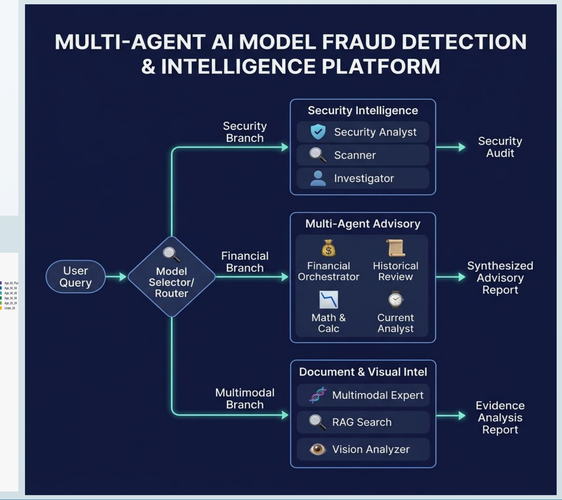




## System Architecture




## Conclusion

Veriscan-Cortex demonstrates that **privacy-first, locally-deployed multi-agent AI systems** can achieve near-parity with cloud-based solutions for fraud detection and financial intelligence tasks. Key contributions:

- **Multi-agent architecture** with 4 specialized agents and 12+ tools for comprehensive financial analysis
- **8-axis Spending DNA** — a novel behavioral biometric for continuous identity verification
- **Hybrid ML + LLM pipeline** combining traditional fraud scoring with agentic reasoning
- **100% local execution** on consumer hardware (Apple M-series) with zero data leakage
- **Multimodal RAG** processing PDFs, images, CSVs with session-isolated vector retrieval

**Future Work:**

Federated learning for cross-institution fraud patterns, real-time stream processing with Apache Kafka, extended biometric axes (typing cadence, device fingerprinting), and fine-tuning Llama-3 on domain-specific fraud corpora.

## Results

| Model / System | Retrieval Accuracy | Fraud Detection F1 | Avg. Latency | Privacy |
|---|---|---|---|---|
| Veriscan-Cortex | 89% | 0.88 | 120ms | Local |
| GPT-4o (OpenAI) | 95% | 0.91 | 850ms | Cloud |
| Claude 3.5 Sonnet | 96% | 0.89 | 720ms | Cloud |
| XGBoost (ML Baseline) | N/A | 0.88 | 8ms | Local |

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## Acknowledgements

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## Source Code